

Calc 1 Lecture 12: Logarithmic differentiation

So far, we've seen:

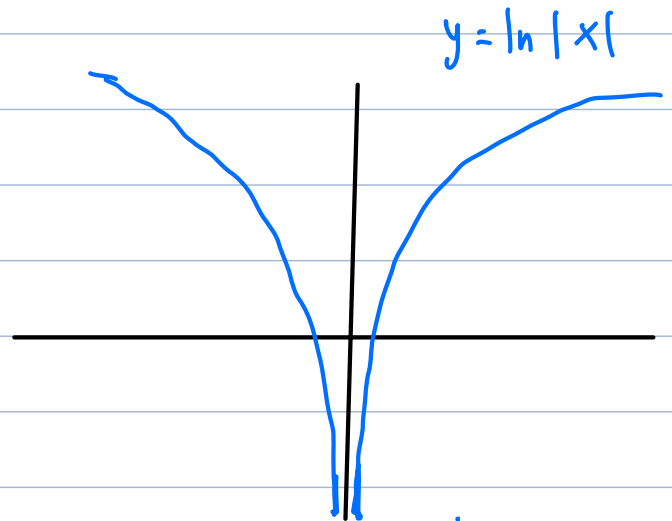
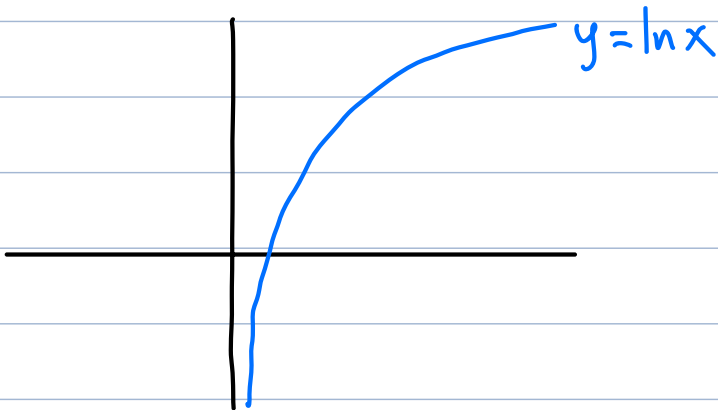
$$\boxed{\frac{d}{dx}(\ln(x)) = \frac{1}{x}}$$

What about $\frac{d}{dx}(\log_a(x))$ for any base a ?

$$\log_a x = \frac{\ln(x)}{\ln(a)} \quad (\text{change of base formula})$$

$$\frac{d}{dx}(\log_a x) = \frac{d}{dx}\left(\frac{\ln(x)}{\ln(a)}\right) = \frac{1}{\ln(a)} \cdot \frac{d}{dx}(\ln x) = \boxed{\frac{1}{\ln(a)} \cdot \frac{1}{x}}$$

Another useful function: $\ln(|x|)$



$$\ln |x| = \begin{cases} \ln x & \text{if } x > 0 \\ \ln(-x) & \text{if } x < 0 \end{cases}$$

$$\Rightarrow \frac{d}{dx} \ln |x| = \begin{cases} \frac{1}{x} & \text{if } x > 0 \\ \frac{1}{-x} \cdot \underbrace{\frac{d}{dx}(-x)}_{-1} & \text{if } x < 0 \end{cases}$$

$\underbrace{\hspace{10em}}_{\frac{1}{x}}$

To summarize:

$$\boxed{\frac{d}{dx}(\ln |x|) = \frac{1}{x}}$$

Observation: Suppose we want to find the derivative of $\ln(f(x))$. We can use the chain rule to get:

$$\frac{d}{dx}(\ln(f(x))) = \frac{1}{f(x)} \cdot f'(x) = \frac{f'(x)}{f(x)} \quad (\text{chain rule})$$

$$\boxed{\frac{d}{dx}(\ln(f(x))) = \frac{f'(x)}{f(x)}}$$

Example: Find $\frac{d}{dx}(\ln(\sin x)) = \frac{\frac{d}{dx} \sin(x)}{\sin(x)} = \frac{\cos(x)}{\sin(x)} = \boxed{\cot x}$
($f(x) = \sin x$)

Example: Find $\frac{d}{dx} \left(\ln \left(\frac{x+1}{\sqrt{x-2}} \right) \right)$

Idea: first simplify using log properties.

$$\ln \left(\frac{x+1}{\sqrt{x-2}} \right) = \ln(x+1) - \underbrace{\ln(\sqrt{x-2})}$$

$$\ln((x-2)^{1/2}) = \frac{1}{2} \ln(x-2)$$

$$= \ln(x+1) - \frac{1}{2} \ln(x-2)$$

Now take derivative:

$$\frac{d}{dx} \ln \left(\frac{x+1}{\sqrt{x-2}} \right) = \underbrace{\frac{d}{dx} (\ln(x+1) - \frac{1}{2} \ln(x-2))}$$

$$\frac{\frac{d}{dx}(x+1)}{x+1} - \frac{1}{2} \frac{\frac{d}{dx}(x-2)}{x-2}$$

$$= \boxed{\frac{1}{x+1} - \frac{1}{2} \cdot \frac{1}{x-2}}$$

We can use this simple observation to find the derivatives of functions even if they do not consist of logarithms.

Example: Use logarithmic differentiation to find

$$\frac{d}{dx} \left(\frac{x^{3/4} \sqrt{x^2+1}}{(3x+2)^5} \right)$$

Idea: $y = \frac{x^{3/4} \sqrt{x^2+1}}{(3x+2)^5}$ $\downarrow$ take $\ln$

$$\ln y = \ln \left(\frac{x^{3/4} \sqrt{x^2+1}}{(3x+2)^5} \right) \downarrow \text{use log properties}$$

$$\ln y = \ln(x^{3/4}) + \ln(\sqrt{x^2+1}) - \ln((3x+2)^5)$$

$$\ln y = \frac{3}{4} \ln(x) + \frac{1}{2} \ln(x^2+1) - 5 \ln(3x+2) \quad \downarrow \frac{d}{dx}$$

$$\frac{d}{dx}(\ln y) = \frac{3}{4} \frac{d}{dx}(\ln x) + \frac{1}{2} \frac{d}{dx} \ln(x^2+1) - 5 \frac{d}{dx} \ln(3x+2)$$

$$\frac{d}{dx}(\ln(f)) = \frac{f'}{f}$$

$$\frac{y'}{y} = \frac{3}{4} \cdot \frac{1}{x} + \frac{1}{2} \cdot \frac{\frac{d}{dx}(x^2+1)}{x^2+1} - 5 \cdot \frac{\frac{d}{dx}(3x+2)}{3x+2}$$

mult.
by y

$$\frac{y'}{y} = \frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2}$$

plug in
y as
function
of x

$$y' = y \cdot \left(\frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2} \right)$$

$$y' = \frac{x^{3/4} \sqrt{x^2+1}}{(3x+2)^5} \left(\frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2} \right)$$

To recap the method of logarithmic differentiation:

Given a function $y = f(x)$, logarithmic differentiation is appropriate if f is a combination of products, quotients and powers of other functions.

To apply the method:

step 1: Take $\ln(y) = \ln(f(x))$ and use laws of logs to simplify the expression on the right.

step 2: Differentiate the above equation implicitly with respect to x .

step 3: solve for y'

*: if $f(x)$ is negative at some points, $\ln(f(x))$ may not be defined. We can salvage this situation by applying an absolute value before taking the log.

$$\ln(|y|) = \ln(|f(x)|)$$

Example: Use logarithmic differentiation to find

$$\frac{d}{dx} \left(\frac{\sin^3(x) \tan(x)}{(x^2+1)^2} \right)$$

$$y = \frac{\sin^3(x) \tan(x)}{(x^2+1)^2}$$

Step 1: take $\ln$ of both sides and simplify

$$\ln(|y|) = \ln\left(\left|\frac{\sin^3(x) \tan(x)}{(x^2+1)^2}\right|\right)$$

$$= \ln\left(\frac{|\sin x|^3 |\tan x|}{(x^2+1)^2}\right)$$

$$= \ln(|\sin x|^3) + \ln|\tan x| - \ln((x^2+1)^2)$$

$$\ln|y| = 3 \ln|\sin x| + \ln|\tan x| - 2 \ln(x^2+1)$$

Step 2: differentiate both sides $\left(\frac{d}{dx} \ln|f(x)| = \frac{f'(x)}{f(x)}\right)$

$$\frac{d}{dx} \ln|y| = 3 \frac{d}{dx} \ln|\sin x| + \frac{d}{dx} \ln|\tan x| - 2 \frac{d}{dx} \ln(x^2+1)$$

$$\frac{y'}{y} = 3 \frac{\frac{d}{dx} \sin x}{\sin x} + \frac{\frac{d}{dx} \tan x}{\tan x} - 2 \frac{\frac{d}{dx} (x^2+1)}{x^2+1}$$

$$\frac{y'}{y} = 3 \frac{\cos x}{\sin x} + \frac{\sec^2 x}{\tan x} - 2 \cdot \frac{2x}{x^2+1}$$

$$\frac{y'}{y} = 3 \cot x + \frac{1}{\sin x \cos x} - \frac{4x}{x^2+1}$$

Step 3: solve for y'

$$y' = y \left(3 \cot x + \frac{1}{\sin x \cos x} - \frac{4x}{x^2+1} \right)$$

$$= \left[\frac{\sin^3(x) \tan(x)}{(x^2+1)^2} \left(3 \cot x + \frac{1}{\sin x \cos x} - \frac{4x}{x^2+1} \right) \right]$$

Example Using logarithmic differentiation to prove the general power rule.

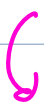
$$\frac{d}{dx} (x^n) = n x^{n-1} \quad (\text{for any number } n)$$

Proof: $y = x^n$

ln



log properties $\ln y = \ln(x^n)$



$\frac{d}{dx}$ $\ln y = n \cdot \ln(x)$



$$\frac{y'}{y} = n \cdot \frac{1}{x}$$

solve for y'



$$y' = n \cdot \frac{y}{x}$$

subst. y as function of x

$$= n \cdot \frac{x^n}{x} = \boxed{n \cdot x^{n-1}}$$

Differentiating exponentials

it helps to classify expressions involving exponentiation into 4 types:

a) $y = \text{constant}$ ^{constant} (e.g. $y = e^{\sin(5)}$)

$$y' = 0$$

b) $y = \text{constant}$ ^{function of x} (e.g. $y = 3^{\sin(x)}$)

$$y' = ?$$

idea: change to base e OR $\frac{d}{du}(3^u) = \ln 3 \cdot 3^u$

let $u = \sin(x)$

$$\Rightarrow \frac{d}{dx}(3^{\sin x}) = \frac{d}{dx}(3^u)$$

$$= \frac{d}{du}(3^u) \cdot \frac{du}{dx}$$

$$= \ln 3 \cdot 3^u \cdot \cos x$$

$$= \boxed{\ln 3 \cdot 3^{\sin x} \cdot \cos x}$$

c) $y = (\text{function of } x)^{\text{constant}}$ (e.g. $y = (\sin(x))^3$)

Can solve using chain rule together with power rule.

e.g. $\frac{d}{dx} (\sin(x)^3)$

$= \underset{\text{chain}}{3 \cdot (\sin(x))^2} \cdot \frac{d}{dx} (\sin(x)) = \boxed{3 \sin^2(x) \cos(x)}$

d) $y = (\text{function of } x)^{\text{function of } x}$ (e.g. $y = x^{\sin(x)}$)

Can solve either using chain rule
by rewriting in base e

or by logarithmic differentiation
recommended approach

Example: $\frac{d}{dx} (x^{\sin x}) = ?$

$\ln$ $y = x^{\sin x}$

$\log$ Properties $\ln y = \ln (x^{\sin x})$

$\ln y = \sin x \cdot \ln(x)$

$\frac{d}{dx}$



$$\frac{y'}{y} = \underbrace{\frac{d}{dx} (\sin x \cdot \ln x)}_{\text{product rule}}$$

$$\underbrace{\frac{d}{dx} (\sin x)} \cdot \ln x + \sin(x) \cdot \underbrace{\frac{d}{dx} (\ln x)}$$

$$\cos x \cdot \ln x + \sin(x) \cdot \frac{1}{x}$$

Solve
for y'



$$\frac{y'}{y} = \cos x \cdot \ln x + \frac{\sin(x)}{x}$$

Plug in y



$$y' = y \left(\cos x \cdot \ln x + \frac{\sin(x)}{x} \right)$$

as
function of x

$$y' = x^{\sin x} \left(\cos x \cdot \ln x + \frac{\sin(x)}{x} \right)$$

Some more examples:

a) Find $\frac{d}{dx} \left((1+\sqrt{x})^{(3x-1)} \right)$

b) Find $\frac{d}{dx} \left(x^{(x^x)} \right)$ (Caution: $x^{(x^x)} \neq (x^x)^x$)

a) $y = (1+\sqrt{x})^{3x-1}$

log properties $\ln \left(\ln y = \ln \left((1+\sqrt{x})^{3x-1} \right) \right)$

$\frac{d}{dx} \left(\ln y = (3x-1) \cdot \ln(1+\sqrt{x}) \right)$

$\left(\frac{d}{dx} \ln(f) = \frac{f'}{f} \right) \quad \frac{y'}{y} = \frac{d}{dx} \left((3x-1) \cdot \ln(1+\sqrt{x}) \right)$

$$= \underbrace{\frac{d}{dx} (3x-1)}_3 \cdot \ln(1+\sqrt{x}) + (3x-1) \cdot \underbrace{\frac{d}{dx} \ln(1+\sqrt{x})}_{\frac{1}{2\sqrt{x}} \left[\frac{d}{dx} (1+\sqrt{x}) \right]}$$
$$= 3 \ln(1+\sqrt{x}) + (3x-1) \cdot \frac{1}{2\sqrt{x}} \cdot \frac{1}{1+\sqrt{x}}$$

Solve for y $\frac{y'}{y} = 3 \ln(1+\sqrt{x}) + (3x-1) \cdot \frac{1}{2\sqrt{x}} \cdot \frac{1}{1+\sqrt{x}}$

plug in y $y' = y \cdot \left(3 \ln(1+\sqrt{x}) + \frac{3x-1}{2\sqrt{x}(1+\sqrt{x})} \right)$

$$= (1+\sqrt{x})^{3x-1} \cdot \left(3 \ln(1+\sqrt{x}) + \frac{3x-1}{2\sqrt{x}(1+\sqrt{x})} \right)$$

b) $y = x^{(x^x)}$

$\frac{d}{dx}$ $\ln y = \ln \left(x^{(x^x)} \right) = \underbrace{(x^x) \cdot \ln(x)}$

$$\frac{y'}{y} = \frac{d}{dx} \left(x^x \cdot \ln(x) \right)$$

$$= \underbrace{\frac{d}{dx} (x^x)} \cdot \ln(x) + x^x \cdot \underbrace{\frac{d}{dx} (\ln(x))}_{\frac{1}{x}}$$

use log diff again!

$\frac{d}{dx}$ $\frac{y}{\ln y} = \frac{y = x^x}{\ln y} = \ln(x^x) = x \ln(x)$

$$\frac{y'}{y} = \frac{d}{dx} (x \ln(x)) = 1 \cdot \ln(x) + x \cdot \frac{1}{x} = \ln x + 1$$

$$\Rightarrow y' = y \cdot (\ln x + 1) = \boxed{x^x \cdot (\ln(x) + 1)} \frac{d}{dx}(x^x)$$

$$\frac{y'}{y} = \underbrace{\frac{d}{dx}(x^x) \cdot \ln(x)}_{x^x \cdot (\ln x + 1)} + x^x \cdot \underbrace{\frac{d}{dx}(\ln(x))}_{\frac{1}{x}}$$

$$y' = y \cdot \left(x^x \cdot (\ln x + 1) \cdot \ln x + x^x \cdot \frac{1}{x} \right)$$

$$= y \cdot x^x \left((\ln x + 1) \cdot \ln x + \frac{1}{x} \right)$$

Subst.
y as
function
of x

$$= \boxed{x^{(x^x)} \cdot x^x \cdot \left((\ln x + 1) \cdot \ln x + \frac{1}{x} \right)}$$

Revisiting e as a limit

Remember: $e = \lim_{x \rightarrow 0} (1+x)^{1/x}$

Justification:

start with $\frac{d}{dx} (\ln(x)) = \frac{1}{x}$

limit def. of derivative

$$\lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln(x)}{h} = \frac{1}{x}$$

at $x=1$:

$$\lim_{h \rightarrow 0} \frac{\ln(1+h) - \overbrace{\ln(1)}^0}{h} = 1$$

$$\lim_{h \rightarrow 0} \underbrace{\frac{\ln(1+h)}{h}} = 1$$

$$\frac{1}{h} \cdot \ln(1+h) = \ln((1+h)^{1/h})$$

$$\lim_{h \rightarrow 0} \ln((1+h)^{1/h}) = 1$$

exponentiate

$$\lim_{h \rightarrow 0} (1+h)^{1/h} = e^1 = e$$